

SAFETY DATA SHEET (GHS)

Phenol | CAS: 108-95-2 | Formula: C₆H₅OH

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA REACH registration dossier — Phenol, CAS 108-95-2 (<https://echa.europa.eu/documents/10162/3e04f30d-9953-4824-ba04-defa32a130fa>)
- PubChem CID 996 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/996>
- US EPA, 'Phenol — Hazard Summary', <https://www.epa.gov/sites/default/files/2016-09/documents/phenol.pdf>
- New Jersey Dept. of Health, Right to Know Hazardous Substance Fact Sheet — Phenol
- Sigma-Aldrich SDS T9424 / P5566 — Phenol
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Phenol

CAS Number: 108-95-2

Chemical formula: C₆H₅OH (C₆H₆O)

Molecular weight: 94.11 g/mol

Synonyms: Hydroxybenzene; Carboic acid; Phenylic acid; Benzenol

Intended use: Process product — CDN decomposer product; downstream distillation (DIST) and final product storage. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Acute Tox. 3 (oral) — H301; Acute Tox. 3 (dermal) — H311; Acute Tox. 3 (inhalation) — H331; Skin Corr. 1B — H314 (also covers serious eye damage); Muta. 2 — H341; STOT RE 2 — H373 (kidney)

Signal word: Danger

Hazard statements (H-codes): H301 (Toxic if swallowed); H311 (Toxic in contact with skin); H314 (Causes severe skin burns and eye damage); H331 (Toxic if inhaled); H341 (Suspected of causing genetic defects); H373 (May cause damage to organs — kidney — through prolonged or repeated exposure)

Precautionary statements (P-codes): P261, P264, P270, P271, P280, P301+P310, P302+P352, P303+P361+P353, P304+P340, P305+P351+P338, P310, P321, P361+P364, P405, P501

GHS pictograms: GHS05 (Corrosion), GHS06 (Skull and crossbones), GHS08 (Health hazard)

Other hazards: Rapid and extensive dermal absorption can cause systemic toxicity (CNS depression, cardiac arrhythmia, hepatic/renal injury) even from relatively small contact areas — among the most acutely dangerous dermal exposure hazards in the plant. Solid below 41°C; requires heat tracing to remain liquid in piping/storage. Not PBT/vPvB.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Composition: Pure substance — Phenol, CAS 108-95-2

Typical product purity: Refined product grade typically >99.5 wt% phenol; trace impurities may include acetophenone, cumene, AMS, water — confirm against PPCL product specification

SECTION 4 — FIRST AID MEASURES

Inhalation: Remove to fresh air immediately. If breathing has stopped, give artificial respiration. Administer oxygen if breathing is difficult. Obtain immediate medical attention.

Skin contact: EMERGENCY — flush immediately and continuously with large amounts of water for at least 15-20 minutes while removing contaminated clothing. Polyethylene glycol (PEG 300/400) may be used as a secondary decontaminant if available and per site procedures, but water flushing must not be delayed. Obtain immediate medical attention — even small splashes can be fatal due to rapid absorption.

Eye contact: Flush with copious water for at least 15-20 minutes, holding eyelids open. Obtain immediate medical/ophthalmological attention.

Ingestion: Do NOT induce vomiting. If conscious, rinse mouth and give water. Obtain immediate medical attention — risk of severe corrosive injury and systemic poisoning.

Notes to physician: Monitor for CNS depression, cardiac arrhythmias, metabolic acidosis, hepatic and renal injury. Activated charcoal may be considered for ingestion per poison control guidance. No specific antidote.

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Water spray/fog, alcohol-resistant foam, dry chemical, carbon dioxide

Unsuitable extinguishing media: High-pressure water jet that may scatter molten/burning phenol

Special hazards: Combustible liquid above flash point (~79°C); molten phenol handling (if heated above MP for storage) presents additional thermal/splash hazard. Combustion produces toxic fumes.

Special protective equipment for firefighters: Full turnout gear with SCBA; chemical-resistant clothing to prevent skin contact with molten/liquid phenol or firewater run-off

Hazardous combustion products: Carbon monoxide, carbon dioxide, and various phenolic decomposition products/smoke

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Evacuate non-essential personnel. Wear full chemical-resistant PPE (Section 8) — splash on skin is a medical emergency. Use SCBA for vapour/mist releases or in confined spaces.

Environmental precautions: Prevent entry to drains, sewers and watercourses — phenol is toxic to aquatic organisms even at low concentrations and imparts taste/odour to water. Contain/bund spills.

Containment and cleanup: For molten/liquid spills, allow to solidify if safe to do so, then collect mechanically; for liquid spills, absorb with inert absorbent (sand, vermiculite). Wash contaminated surfaces with large amounts of water. Collect in labelled containers for disposal.

Reference to other sections: See Section 4 (First Aid — dermal exposure is an emergency), Section 8 (PPE) and Section 13 (Disposal)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Avoid all skin and eye contact — use closed transfer systems, splash-proof PPE and eyewash/safety shower stations within immediate reach of all phenol handling points. Avoid generation of mist/vapour; provide local exhaust ventilation.

Storage conditions: Store in heat-traced/insulated tanks and piping maintained above the melting point (41°C) to keep phenol molten and prevent line plugging from solidification; alternatively store and handle as a solid below 40°C with appropriate handling — PPCL design basis maintains product molten (~45-60°C typical).

Temperature limits: Keep above 41°C (melting point) if maintained in liquid state to avoid solidification/blockage of lines and instrumentation; avoid prolonged exposure above ~150-160°C where thermal degradation/colour formation can occur

Pressure: Atmospheric to low pressure storage typical; N₂ blanketing may be used to limit oxidation/colour formation and moisture ingress

Segregation requirements: Segregate from strong oxidisers (e.g. concentrated nitric acid), strong bases, and formaldehyde/aldehydes (exothermic condensation/resin-forming reactions)

Special requirements: Heat tracing with high/low temperature alarms on all phenol lines and storage to prevent freeze-plugging (low temp) and degradation (high temp); dedicated eyewash/deluge showers at all sampling and connection points

Specific end-use recommendations: Maintain product quality (colour) by minimising residence time at elevated temperature and excluding oxygen/moisture ingress where practical

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): ACGIH TLV-TWA: 5 ppm (19 mg/m³), Skin notation (absorption contributes significantly to overall exposure). OSHA PEL: 5 ppm (19 mg/m³) TWA, skin. Thailand OEL: aligned with ACGIH 5 ppm TWA — confirm current Ministerial Notification value.

TWA: 5 ppm (19 mg/m³) — ACGIH/OSHA

STEL: Not separately established by ACGIH (TWA value applies); some jurisdictions specify a STEL — confirm locally

Ceiling: Not established

Skin notation: Yes — critical exposure pathway

IARC/NTP/OSHA carcinogen status: IARC Group 3 (not classifiable as to carcinogenicity to humans); not listed as a carcinogen by NTP or OSHA

Engineering controls: Closed systems, local exhaust ventilation at sampling/loading points, continuous heat tracing with temperature monitoring

PPE — Respiratory: Full-facepiece air-purifying respirator with organic vapour cartridge for routine tasks with potential vapour exposure; SCBA for emergency response

PPE — Gloves: Phenol-resistant gloves — butyl rubber or laminate (e.g. Silver Shield/4H-type) recommended; nitrile and natural rubber offer limited/short breakthrough times and are NOT recommended for phenol

PPE — Eyes: Chemical splash goggles plus face shield for any task with splash potential

PPE — Body: Chemical-resistant suit (e.g. Tyvek/Tychem with taped seams) for tasks with splash potential; heat-resistant variant where molten phenol is handled

Biological monitoring: Urinary phenol can be used as a biomarker of exposure (with consideration of dietary background levels) — consult occupational health physician for biological exposure index (BEI) guidance

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: White to light pink crystalline solid below 41°C; colourless to light pink liquid above 41°C — colour can darken (pink/red) on exposure to light, air or trace iron contamination

Odour: Sweet, sickly, 'medicinal'/disinfectant odour

Odour threshold: ~0.04-0.06 ppm (very low — odour detectable well below toxic levels)

pH: Aqueous solution mildly acidic (pKa ~10, but solutions of phenol in water are weakly acidic, pH ~5-6 for saturated solution)

Melting point / Freezing point: 40.5°C — CRITICAL for heat tracing/line design (solidifies and can block lines/instrumentation below this temperature)

Boiling point (at 1 atm): 181.7°C

Flash point: 79°C (closed cup)

Evaporation rate: Low (relative to water)

Flammability (solid/gas): Combustible solid/liquid — not classified as readily flammable (Flash point >60°C)

Flammability limits in air: LEL ~1.3% v/v; UEL ~9.0% v/v (approximate, vapour-phase)

Vapour pressure (at 20°C): ~0.05 kPa (~0.4 mmHg) at 20°C; ~0.2 kPa at 40°C (above MP)

Vapour density (air = 1): ~3.2 (heavier than air)

Relative density (water = 1): ~1.07 (liquid at ~25°C, supercooled/molten)

Solubility (water): ~8.3 g/100 mL at 20°C (moderately soluble; fully miscible above ~68°C)

Partition coefficient (log Kow): 1.46

Auto-ignition temperature: ~715°C

Decomposition temperature: Thermally stable under normal process conditions; decomposition/charring above ~400°C producing CO, CO₂ and phenolic fume

Viscosity: ~3.4 mPa·s at 50°C (molten)

Explosive properties: Not classified as explosive

Oxidising properties: Not an oxidiser

Molecular weight: 94.11 g/mol

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Reacts with strong oxidisers (can be vigorous/exothermic) and with formaldehyde/other aldehydes (exothermic condensation, can form resins under uncontrolled conditions)

Chemical stability: Stable under normal storage and handling conditions when kept within the recommended temperature range and protected from strong oxidisers

Possibility of hazardous reactions: Exothermic reaction/polymerisation with formaldehyde (phenol-formaldehyde resin chemistry) if uncontrolled contact occurs; violent reaction with strong oxidising agents (e.g. concentrated nitric acid, chromic

acid)

Conditions to avoid: Prolonged exposure to air/light (discolouration), contamination with iron or other catalytic metal ions (colour formation), temperatures below 41°C in lines without heat tracing (solidification/blockage)

Incompatible materials: Strong oxidisers, strong bases (forms phenolate salts — exothermic), aldehydes (especially formaldehyde), amines, calcium hypochlorite

Hazardous decomposition products: Carbon monoxide and carbon dioxide on combustion; phenolic fume

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Dermal (major concern — rapid absorption), inhalation, ingestion, eye contact

Acute toxicity — Oral LD50: ~317 mg/kg, rat

Acute toxicity — Dermal LD50: ~850 mg/kg, rabbit (note: death can occur from dermal exposure to relatively small surface areas due to rapid systemic absorption)

Acute toxicity — Inhalation LC50: ~316 mg/m³ (4 h), rat (approximate, literature range — confirm with manufacturer data)

Skin corrosion/irritation: Corrosive — causes severe burns; numbness at the contact site can mask the severity of injury

Serious eye damage/irritation: Causes serious eye damage

Respiratory/skin sensitisation: Not classified as a sensitiser

Germ cell mutagenicity: Suspected of causing genetic defects (H341, Muta. 2) based on in-vitro/in-vivo genotoxicity studies

Carcinogenicity: IARC Group 3 (not classifiable); not listed by NTP/OSHA

Reproductive toxicity: Not classified, though some developmental effects reported at high doses in animal studies — not sufficient for classification

STOT (single exposure): CNS depression, cardiac arrhythmia and respiratory effects reported following acute high-level exposure

STOT (repeated exposure): H373 — kidney damage with repeated/prolonged exposure

Aspiration hazard: Not classified

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC50/EC50): Fish LC50 (96h) in the range of ~5-20 mg/L for various species (e.g. fathead minnow); toxic to aquatic organisms and imparts taste/odour to water at very low concentrations (sub-mg/L)

Persistence and degradability: Readily biodegradable under aerobic conditions

Bioaccumulative potential: Low (log K_{ow} 1.46) — not expected to bioaccumulate significantly

Mobility in soil: Moderate to high mobility; soluble in water

Other adverse effects: No data indicating ozone-depletion potential; avoid release to watercourses due to taste/odour and toxicity thresholds at low concentrations

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Dispose of via licensed hazardous waste contractor — typically incineration with appropriate emission controls. Do not discharge to drains or watercourses.

Contaminated packaging: Empty containers retain residue/vapour — handle as hazardous waste; do not reuse

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535 and Factory Act waste management requirements

SECTION 14 — TRANSPORT INFORMATION

UN Number: UN 1671 (Phenol, solid) or UN 2312 (Phenol, molten) — select per shipping state

Proper shipping name: PHENOL, SOLID or PHENOL, MOLTEN

Transport hazard class(es): Class 6.1 (Toxic substance)

Packing group: II

Environmental hazards (marine pollutant): Not classified as a marine pollutant under current GHS aquatic classification, but local discharge limits apply

Special precautions for user: Molten phenol transport requires heated/insulated tank containers maintained above 41°C

Bulk transport (MARPOL/IBC): Classify per IBC Code if shipped in bulk by sea

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 108-95-2)

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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